Building Materials

Wei, Ya （魏亚）
Department of Civil Engineering

Tel. : +86-10-62771646
Email: yawei@tsinghua.edu.cn

Chapter One Introduction
Chapter Two The Fundamentals
Chapter Three Metal
Chapter Four Concrete
Chapter Five Asphalt
Chapter Six Masonry
Chapter Seven Wood

4.7 Volume Change

Thermal Deformation

- Concrete expands slightly as temperature rises and contracts as temperature falls.
- An average value for the coefficient of thermal expansion of concrete (CTE) is about 10 millionths per degree Celsius.
- Aggregate type has the greatest influence on CTE.

Aggregate type (from one source)	Coefficient of expansion, millionths per °C	Coefficient of expansion, millionths per °F
Quartz	11.9	6.6
Sandstone	11.7	6.5
Gravel	10.8	6.0
Granite	9.5	5.3
Basalt	8.6	4.8
Limestone	6.8	3.8

4.7 Volume Change

Thermal Deformation

The magnitude of the thermal stress is directly proportional to the magnitude of the temperature change that concrete is exposed to:

$$\text{Thermal Stress} = \sigma_T = \Delta T \cdot \alpha_t \cdot E_c \cdot K_r$$

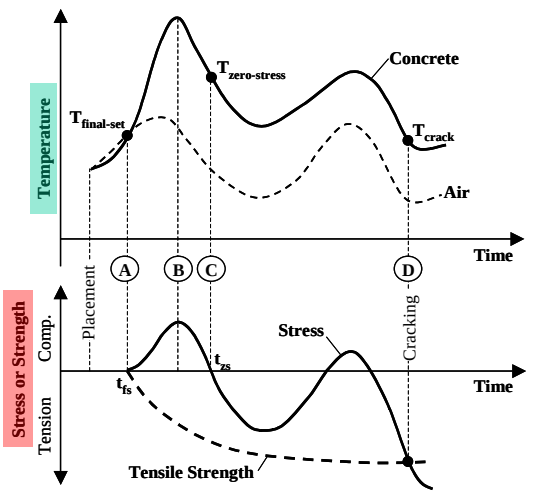
where,

ΔT = Temperature Change
α_t = Coefficient of Thermal Expansion (strain/°C)
E_c = Creep adjusted Modulus of Elasticity
K_r = Degree of restrain factor

4.7 Volume Change

Thermal Deformation

the development of thermal stresses over time for freshly placed concrete

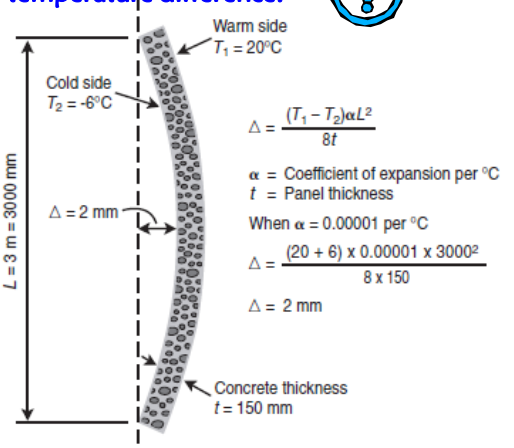


4.7 Volume Change

Thermal Deformation

Curling in a wall panel due to temperature difference:

Precast wall panels and slabs and pavements on ground are susceptible to bending and curling caused by temperature gradients that develop when concrete is cool on one side and warm on the other.



$$\Delta = \frac{(T_1 - T_2)\alpha L^2}{8t}$$

α = Coefficient of expansion per °C
t = Panel thickness

When $\alpha = 0.00001$ per °C

$$\Delta = \frac{(20 + 6) \times 0.00001 \times 3000^2}{8 \times 150}$$

$\Delta = 2$ mm

Chapter Four：Concrete

1. Introduction
2. Compositions
3. Structure
4. Strength and Failure
5. Mix Proporationing
6. Fresh and Early-age Properties
7. Volume Changes
8. **Durability**

4.8 Durability

Definition by ACI Committee 201

Durability is the ability of materials to resist *weathering actions, chemical attack, abrasion* and *other deterioration process*.

Durable concrete will retain its original form, quality, and serviceability when exposed to its environment.

4.8 Durability

Aspects of Durability Study

- corrosion of reinforcing bar
- alkali-aggregate reaction
- carbonation、fire damage
- pore structure – permeability、frost resistance

4.8 Durability

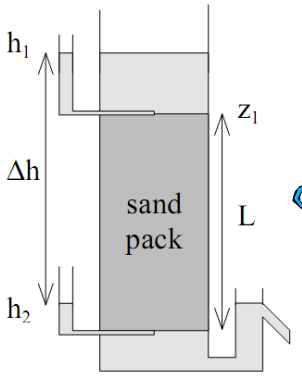
Water

- Water is the cause of many types of physical processes of degradation.
- As a vehicle for transport of aggressive ions, water can also be a source of chemical processes of degradation.
- The physical-chemical phenomena associated with water movements in porous solids are controlled by the permeability of the solid.

4.8 Durability

Permeability (Darcy's Law)

Darcy's experiment was performed to design a filter large enough to ensure the daily requirement of water for the city of Dijon (1856).

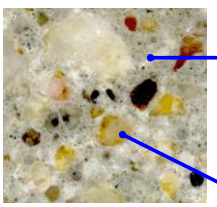


$$q = KA\frac{(h_1 - h_2)}{L}$$

q: volumetric rate [m^3/s]
K: coefficient of permeability
A: Cross-sectional area of sand pack [m^2]
h: water pressure head [m]
L: length of sand pack [m]

4.8 Durability

Permeability



Cement matrix

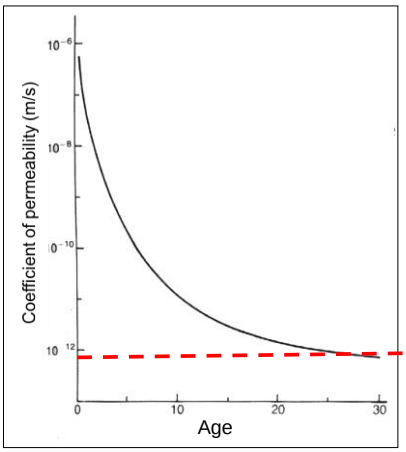
Coarse aggregate



❑ The mixing water is indirectly responsible for permeability of the hydrated cement paste because its content determines first the total space and subsequently the unfilled space after the water is consumed by either cement hydration reactions or evaporation to the environment.

4.8 Durability

Permeability



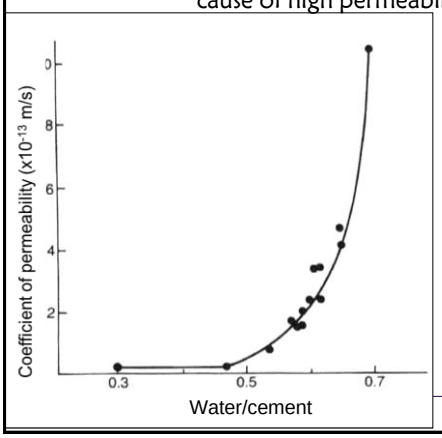
- With the progress of hydration the capillary porosity decreases, the coefficient of permeability decreases.
- However, a further decrease in the porosity from 30 percent to 20 percent would bring about only a small drop in permeability.

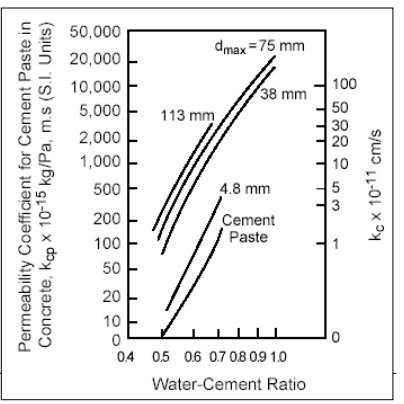
Reason: large pores are reduced in size and number. and there is creation of tortuosity.

4.8 Durability

Permeability

- Reduction in the volume of large (e.g., > 100 nm) capillary voids in the paste matrix would reduce the permeability.
- Microcracking in the transition zone appears to be a major cause of high permeability of concrete in practice.

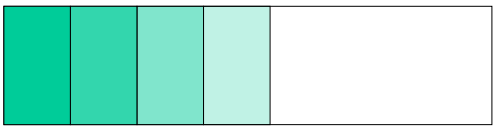




4.8 Durability

Diffusion

Phenomenon of material transport by atomic or particle transport from region of high to low concentration.



Fick's law:

$$J = -D\frac{\partial C}{\partial x}$$

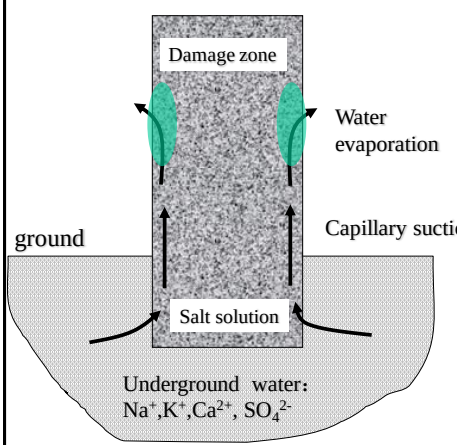
J: is flux (mol/m^2/s)

D: is the diffusion coefficient or diffusivity, m^2/s

$\frac{\partial C}{\partial x}$: is concentration gradient (mol/m^3/m)

4.8 Durability

Salt Crystallization (pure physical action)



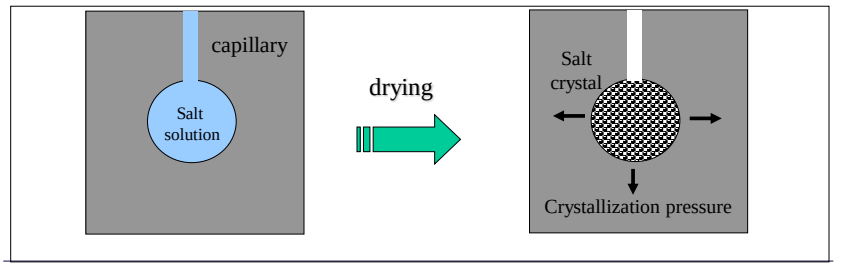




4.8 Durability

Salt Crystallization (pure physical action)

- The crystallization of salts in the pores of concrete can produce stresses that may damage the concrete structure. This can happen when the concentration of the solute (C) exceeds the saturation concentration (C_s).
- Higher C/C_s ratio (degree of supersaturation) produces higher crystallization pressure. NaCl crystallizing at 25C produces 605 atm pressure, strong enough to disrupt most rocks.



4.8 Durability

Deterioration by frost action

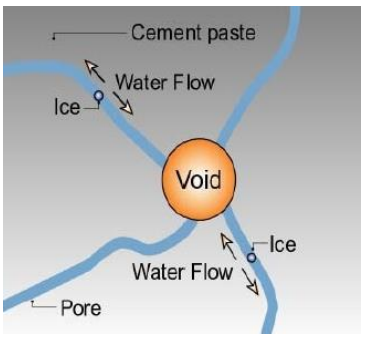




4.8 Durability

Deterioration by frost action

The transformation of ice from liquid water generates a volumetric dilation of 9%. If the transformation occurs in small capillary pores, the ice crystals can damage the cement paste by pushing the capillary walls and by generating hydraulic pressure.

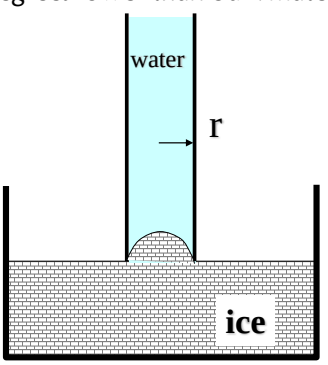


❑ **Hydraulic pressure depends on:**
(a) rate at which ice is formed
(b) permeability of the material
(c) distance to an "escape boundary

4.8 Durability

Deterioration by frost action

Melting-point depression is reduction of melting point of a material with reduction of its size. Nanoscale materials melt at temperatures hundreds of degrees lower than bulk materials.



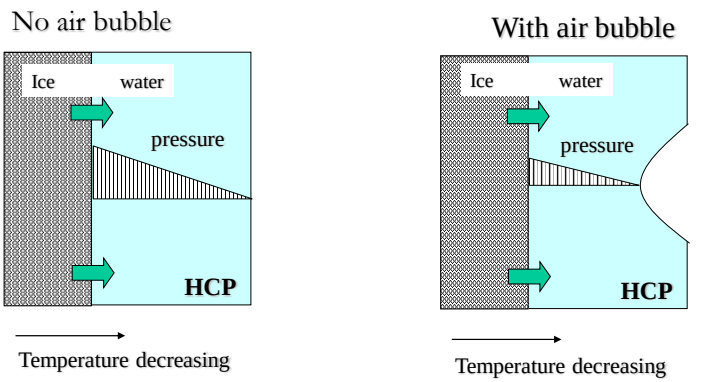
Thomson-Gibbs equation:

Melting-point depression: $\Delta\theta = \frac{2\gamma}{s_f r}$

γ: interface tension
r: Radius of pore
s_f: Heat of fusion

4.8 Durability

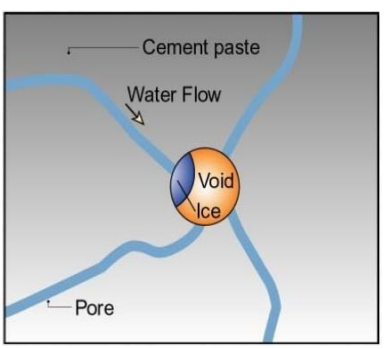
Deterioration by frost action

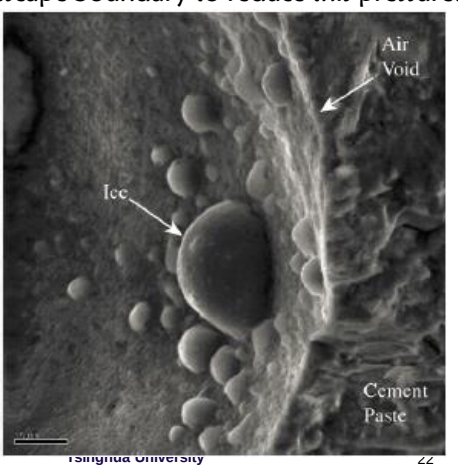


4.8 Durability

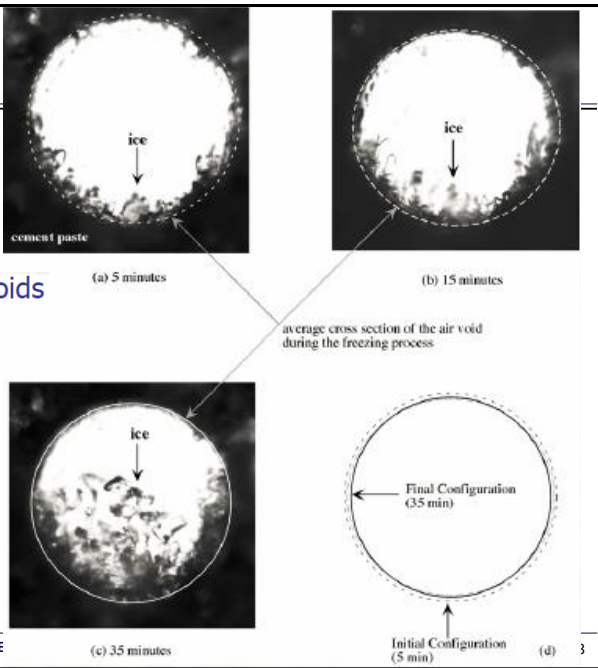
The solution

Air voids can provide an effective escape boundary to reduce this pressure.





Ice Forming in Air Voids



4.8 Durability

Deterioration by frost action



Freeze-thaw resistant concrete should have 4-8% air-bubbles by volume and the air-bubbles should be well distributed and have a distance between each other of less than 0.2 mm in the cement paste matrix (**air-entraining agent**).



4.8 Durability



High Temperature

Concrete probably has the best fire resistance of all common building material. When exposed to fire it does not emit toxic fumes, smoke or partly melt.

- Concrete has a relatively low thermal conductivity and high specific heat and is a good protection for steel against fire.
- Fire can result in 'spalling' of concrete (i.e., breaking of layers or pieces from the concrete surface), due to (i) aggregate expansion, (ii) gas pressure build-up, or (ii) phase transitions.
- Fire may causes extreme initial drying and shrinkage and thereby cause shrinkage cracking by self-restraint or substrate-restraint.

4.8 Durability

Fire in tunnels

Form the standpoint of fire safety of steel structures, a 50- to-100 mm coating of concrete or any other fire-resisting material is routinely specified by building codes.





4.8 Durability

Deterioration by fire

Concrete is able to retain sufficient strength for a reasonably long time.

Effect of temperature on the cement paste

- ❑ degree of hydration
- ❑ moisture state
- ❑ de-hydration:

ettringite > 100^{0}C
$Ca(OH)_2$ 500-600^{0}C
CSH ~ 900^{0}C

Effect of High Temperature on the Aggregate

Siliceous quartz:
573 ^{0}C → sudden volume change

Carbonate:
$MgCO_3$ > 700 ^{0}C, $CaCO_3$ > 900 ^{0}C

4.8 Durability

Chemical Reaction:

Chemical reaction that involve formation of expansive products in hardened concrete can lead to certain harmful effects.

The four phenomena associated with expansive chemical reactions are:

- Sulfate attack
- Alkali-aggregate attack
- Carbonation
- Corrosion of steel in concrete.

4.8 Durability

Types of chemical reaction in concrete affecting durability:

- Reactions with larger products than the reactants, causing expansion and as a result cracking.
- Dissolution (leaching) or other chemical reactions, leading to higher porosity and therefore lower strength.

Important factors are:

(i.) rate of diffusion or permeation;
(ii.) chemical reaction rate.

4.8 Durability

Sulfate Attack

Sulfate attack is a chemical breakdown mechanism where sulfate ions attack components of the cement paste. Sulfate attack can be 'external' or 'internal':

External: due to penetration of sulfates in solution, in groundwater For example, into the concrete from outside.

Internal: due to a soluble sulfate source being incorporated into the concrete at the time of mixing, for example added to cement (to Control setting) can cause sulfate attack.



4.8 Durability

Sulfate Attack The chemistry:

Calcium hydroxide and alumina-bearing phases of hydrated portland cement are most vulnerable to attack by sulfate ions.
i.) Expansive reaction (stress-formation):

$$C_3A + CaSO_4.2H_2O \rightarrow C_4A\bar{S}H_{12}\ (\text{monosulfate}) \rightarrow C_6A\bar{S}_3H_{32}\ (\text{ettringite})$$

ii.) Depending on the cation type associated with the sulfate solution (Na^+ or Mg^{2+}), both $Ca(OH)_2$ and C-S-H present in the hydrated portland cement paste may be converted to gypsum by sulfate attack:

$$Na_2SO_4 + Ca(OH)_2 + 2H_2O \rightarrow CaSO_4.2H_2O + 2NaOH \quad \textbf{or:}$$

$$MgSO_4 + Ca(OH)_2 + 2H_2O \rightarrow CaSO_4.2H_2O + Mg(OH)_2 \quad \textbf{and:}$$

$$3MgSO_4 + 3C\text{-}S\text{-}H + 8H_2O \rightarrow 3CaSO_4.2H_2O + 3Mg(OH)_2 + 2SiO_2.H_2O$$

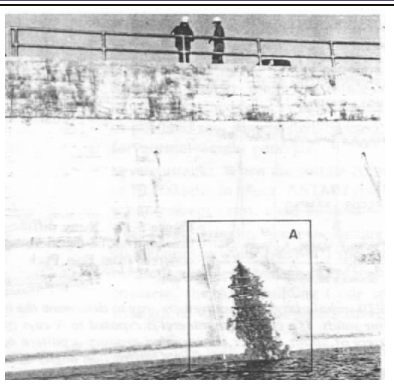
decalcification and destruction of C-S-H

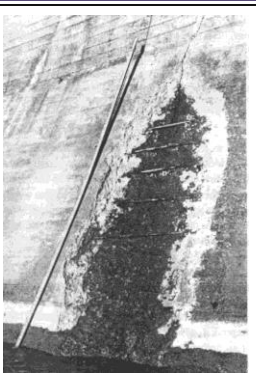
4.8 Durability

Factors influencing sulfate attack:

- amount and nature of the sulfate present,
- level of the water table and its seasonal variation,
-flow of groundwater and soil porosity,
- quality (porosity) of concrete.
- cement content: high sulfate, high alkali, high C_3A

4.8 Durability



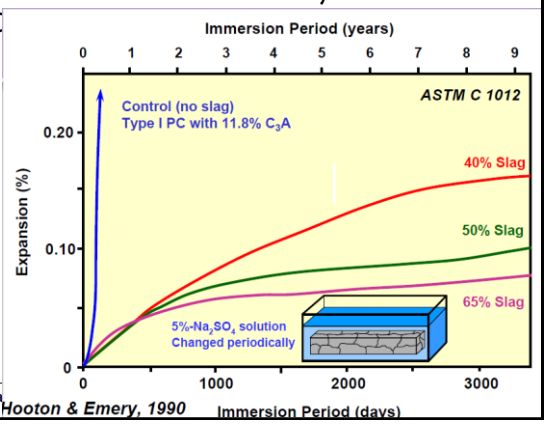


- The quality of concrete, specifically a low permeability, is the best protection against sulfate attack.
- Portland cement containing less than 5 percent C_3A is sufficiently sulfate resisting under moderate conditions of sulfate attack.

4.8 Durability

Sulfate Resistance Of Concrete

- Limits on C_3A
- Use of Supplementary Cementitious Materials
- Limits on W/CM with implied limits on "Permeability"
- Proper Compaction and C



Hooton & Emery, 1990

4.8 Durability

Effect of concrete moisture conditions on carbonation?

Carbonation

- $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

Optimal conditions for carbonation occur at a RH of 50% (range 40-90%)

- Accompanied by shrinkage → carbonation shrinkage
- Makes the steel vulnerable to corrosion (due to reduced alkalinity)



Concrete exposed to CO_2

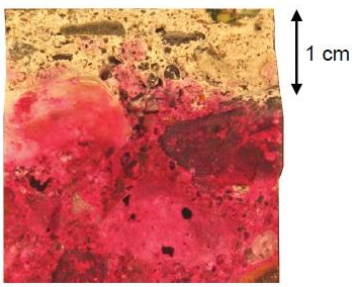


Carbonated concrete

4.7 Volume Change

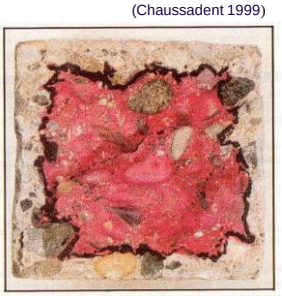
Carbonation

Carbonation test:
Spray a 1% solution of Phenolphthalien on to a fresh fracture concrete surface (or the outside of a concrete core). Noncarbonated areas turn red or purple, carbonated areas stay colorless.



4.8 Durability

(Chaussadent 1999)





Carbonation depth

Experimental model

$$x_{CO_2} = \gamma\, f(h) k \sqrt{t}$$

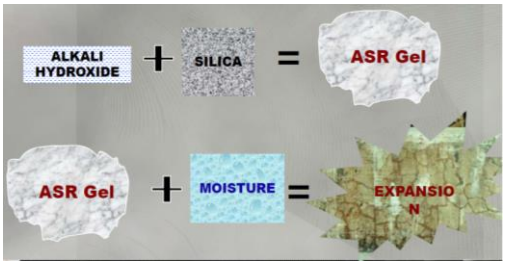
x_{CO2}	Depth of carbonation, [cm]
γ	Concentration of CO_2 on concrete surface, [-]
f	Moisture factor, [-]
h	Concrete humidity, 0 ~ 100% [-]
k	Diffusion coefficient of CO_2 in concrete
t	time, [year]

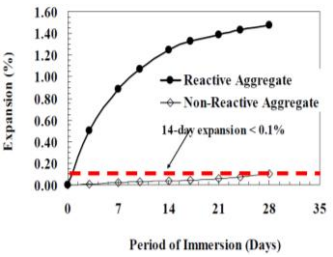
4.8 Durability

Alkali-silica reaction:









4.8 Durability

Alkali-silica reaction:

The chemistry is simple

1) The high pH in the cement paste promotes the hydrolysis of silica:
Si-O-Si + H OH → Si-OH+ Si-OH
aggregate paste

Breakdown of the silica structure by hydroxyl ions

2) Si-OH react with the paste to form Si-O-

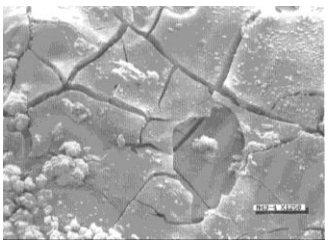
3) Si-O-, adsorbs Na, K, and Ca to form a gel.

Adsorption of alkali ions on new product

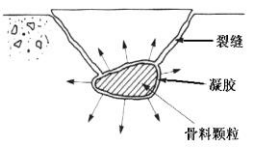
4) This alkali-silicate gel swells in presence of water through the process of osmosis

4.8 Durability

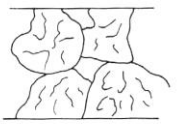
Alkali-silica reaction:







爆皮



表面网状裂缝



4.8 Durability

Alkali-silica reaction:

Factors influencing the reaction

- Alkali content of cement and other sources
- Amount, size and reactivity of alkali-reactive material present in aggregate
- Availability of moisture

Measures for prevention

- Low alkali content cement and mildly reactive aggregate
- Sweetening of aggregate using limestone
- Control of access of water to concrete
- Replacing part of cement by pozzolanic admixtures

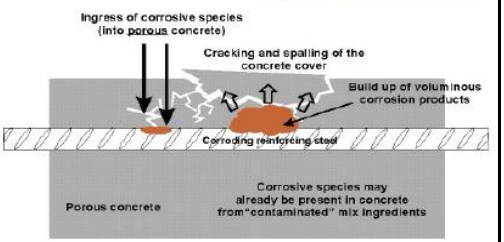
4.8 Durability

Steel Corrosion:









4.8 Durability

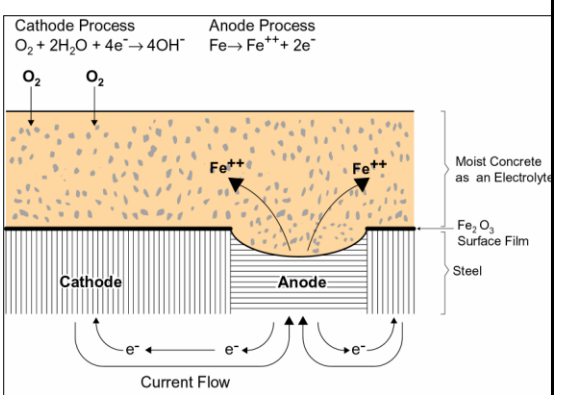
Requirements for Rebar Corrosion

1. **Anode** (location where metal is lost) pH < 11 and/or chloride ion initiate corrosion

2. Cathode (steel surface) Oxygen and water are consumed

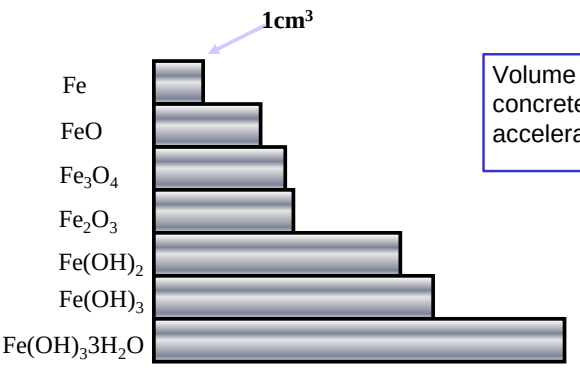
3. Electrical connection (steel) Electron transfer from anode to cathode

4. Electrolytic connection (moist concrete) Ionic transfer from cathode to anode



4.8 Durability

Volume change during corrosion



Volume expansion causes concrete cracking which accelerates corrosion。



QUESTIONS?